



Influence of silicon carbide particulate reinforcement on quasi static and cyclic fatigue fracture behavior of 6061 aluminum alloy composites

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Abstract

In this paper, the quasi-static and cyclic fatigue fracture behavior of aluminum alloy 6061 discontinuously-reinforced with fine particulates of silicon carbide are presented and discussed. The discontinuous particulate-reinforced 6061 aluminum alloy was cyclically deformed to failure at ambient temperature under stress-amplitude controlled conditions. The influence of volume fraction of particulate reinforcement on high cycle fatigue response is presented. The underlying mechanisms governing the fracture behavior during quasi-static and cyclic fatigue deformation are discussed and rationalized in light of concurrent and mutually interactive influences of composite microstructural features, deformation characteristics of the metal matrix and reinforcement particulate, nature of loading and ductility of the microstructure. © 2002 Elsevier Science B.V. All rights reserved.

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1. Introduction

A sustained interest emerging from the need to improve fuel economy, reduce vehicle emissions, while concurrently increasing styling options, improving overall performance, maintaining safety, enhancing quality, assuring reliability, and ensuring profitability, are challenges that need to be addressed by materials used in a spectrum of high performance applications. These requirements culminated in the development and emergence of the 'newer' generation of materials, namely, metal matrix composites (MMCs) [1–4]. Their exceptionally high specific modulus, strength to-weight ratios, fatigue strength, wear resistance, and tailorable properties such as thermal expansion coefficient are few of the important attributes that have made MMCs an attractive and viable candidate for use in weight-sensitive and stiffness-critical components in transportation systems [5–8]. However, the inadequate fracture tough-

ness, limited damage tolerance, poor tensile ductility, inferior fracture-related properties, size limitations on available product forms, and intrinsic material variability, have been a severe limitation to the wide spread use of these composites in performance critical applications [9–14].

A class of MMC materials which is grown to gain increased attention and use in a spectrum of defense, automotive and even consumer-related markets, are those based on aluminum alloy metal matrices and discontinuously-reinforced with particulate, whisker or short-fiber reinforcements. These composites are referred to as DRA-MMCs [6,7]. The presence of a discontinuous reinforcement phase in a continuous aluminum alloy metal matrix can result in properties not attainable by other means, thereby extending the potential range of possible applications [9]. It is possible to systematically engineer improvements in structural efficiency, high specific stiffness, high strength, wear resistance, reliability, high temperature stability, fatigue resistance, crack growth characteristics, and creep-rupture properties. Such improvements would also ensure

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achieving the desired physical properties, such as coefficient of thermal expansion (CTE), by selecting an appropriate discontinuous phase for reinforcing a metal matrix. The improvements in properties are dependent on the mutually interactive influences of: (a) intrinsic properties of the composite constituents; and (b) size, shape, orientation, volume fraction, and distribution of the reinforcing phase in the metal matrix [1–4,7,15–20].

Selection of reinforcement type and geometry (shape) is critical to obtaining the best optimum combination of properties at a substantially low cost [19]. Silicon carbide particulates (SiC_p) are the most preferred reinforcements because enhanced properties are achievable with little or no density penalty [21,22]. For example, incorporation of discontinuous particulate reinforcements in a ductile aluminum alloy metal matrix has resulted in a 15–40% increase in strength and a 30–70% increase in stiffness, compared to the unreinforced counterpart [23,24]. The increase in strength is often more pronounced at elevated temperatures [4,7]. Improvements in elastic modulus of up to 100% have been reported for an aluminum alloy discontinuously-reinforced with 40-vol.% of silicon carbide [22,25]. From a design perspective, the attractiveness in preferring and choosing a discontinuously-reinforced aluminum alloy MMC for many applications stems from an improvement in specific modulus, that is, the density compensated increase in elastic modulus. In particular, the discontinuously reinforced metal matrices have properties that are near isotropic in the three orthogonal directions coupled with improvements in fracture-related properties when compared to the continuously reinforced metal matrices. Furthermore, the discontinuously reinforced metal matrices maintain their amenability to conventional metallurgical processing, fabrication and characterization methods currently used for the unreinforced counterparts [21,22].

Many practical applications for the DRA-MMCs involve cyclic loading and therefore, the fatigue properties and resultant fracture characteristics are of need and interest. While several studies have focused on understanding the influence of discontinuous SiC particulate reinforcement phase on the fatigue crack growth and fracture toughness characteristics of DRAMMCs [25–34], there have been only a few studies to systematically and comprehensively rationalize the influence of reinforcement volume fraction and size on cyclic fatigue response and final fracture behavior [33–43]. Complex relationships can exist between the cyclic fatigue properties and fracture characteristics of a

discontinuous particulate-reinforced aluminum alloy-based MMC. These are dependent upon the interactive influences of: (a) the intrinsic properties of the matrix (composition, aging, microstructure); (b) intrinsic properties of the particulate reinforcement phase (composition, morphology, size, volume fraction and distribution); (c) the extrinsic influence of secondary processing on microstructure; and (d) the influence of test parameters, such as: (1) nature, type and frequency of loading; and (2) cyclic microplasticity.

The objective of this paper is to record the influence of reinforcement volume fraction on the quasi-static and high-cycle fatigue properties and final fracture behavior of an Al–Mg–Si alloy based MMC. Emphasis was given to using an MMC having the same matrix composition and aging condition, thereby, the matrix microstructure and reinforcement particle characteristics, that is: type, size distribution, shape, and interfacial characteristics are held the same, while varying only the volume fraction of the particulate (SiC_p) reinforcement phase. The high cycle fatigue properties and underlying mechanisms governing quasi-static and cyclic fracture behavior of the composite are discussed in light of the mutually interactive influences of intrinsic composite microstructural effects, nature and magnitude of loading, matrix deformation characteristics, and fatigue life.

2. Materials

The MMC materials selected for investigation was based on aluminum alloy 6061 discontinuously-reinforced with particulates of silicon carbide (henceforth referred to as SiC_p). In fact, aluminum alloy 6061 is a popular choice and used fairly extensively as the matrix for many aluminum alloy based MMCs. The primary reasons for selecting 6061 are its acceptable mechanical properties coupled with the relative ease with which it can be extruded [44]. Other important yet contending factors include castability, market acceptance and relative ease of development. The nominal chemical composition (in wt.%) of the matrix alloy is given in Table 1. The particulate reinforced aluminum alloys were produced according to the DURALCAN™ process [45]. The molten composite was cast as a billet and extruded as rods with a cross-section of 17×17 mm. The extruded rods were heat treated at 560 °C for 0.5 h followed by quenching in water and artificial aging at 160 °C for 8 h to get the T6 temper. The 6061/ SiC_p /xvp (where xvp denotes the volume fraction of the SiC_p reinforcement phase) composite materials were provided by DURALCAN USA (San Diego, CA) in the T6 condition. Two different volume fractions of the reinforcing SiC_p phase: 10 and 15-vol.% were considered.

Table 1
Nominal chemical composition of aluminum alloy 6061-T6 (in wt.%)

Element (%)	Si	Fe	Cu	Mn	Mg	Cr	Al
	0.60	0.70	0.30	0.28	1.00	0.21	Reminder

3. Experimental

3.1. Specimen preparation

Test specimen configuration for the quasi-static and cyclic fatigue tests was in accordance with specifications in ASTM Standard E-8 [46]. Blanks of length 150 mm were cut from the as-received composite extrusion using a diamond-coated saw blade. The test specimens were precision machined from the blanks using a diamond-tipped cutting tool. The specimens, machined with stress axis parallel to the extrusion direction, were smooth and cylindrical in the gage section, which measured 6.25 mm in diameter and 25 mm in length. The length-to-diameter ratio of the specimen was chosen so as to ensure that it would not buckle during stress controlled fatigue tests. To minimize the effects of surface irregularities and finish, the machined surface of the test specimens were mechanically ground on a lathe and subsequently hand polished to mirror finish using finer grades of silicon carbide impregnated emery paper to remove all circumferential scratches and surface machining marks.

3.2. Initial microstructure characterization

Metallographic samples were cut from the as-received composite extrusion, mounted in bakelite and mechanically ground and coarse polished on 300, 400, and 600 grit silicon carbide (SiC) impregnated emery paper using copious amounts of water as lubricant. Subsequently, the specimens were fine polished on SiC impregnated emery paper using one- μm alumina-based polishing compound suspended in distilled water as the lubricant. Finish polishing to a mirror-like surface was achieved using 1 μm diamond paste. The morphology and distribution of the reinforcement phase in the aluminum alloy metal matrix and other intrinsic microstructural features were examined in an optical microscope and photographed using bright-field illumination technique.

3.3. Mechanical testing

All mechanical tests were performed on a fully automated, closed-loop, servohydraulic structural test machine (INSTRON: Model 8021) equipped with a 100 KN load cell. A precision alignment was given to the load train prior to the initiation of testing to minimize grip-induced bending. Tensile tests were carried out at a strain rate of 10^{-4} s^{-1} . Stress amplitude-controlled high-cycle fatigue tests were conducted in laboratory air (Relative Humidity of 55%) at ambient temperature ($T = 27^\circ\text{C}$). The fatigue tests were conducted at constant cyclic frequency of 5 Hz and at a stress ratio [$R = (\sigma_{\text{minimum}})/(\sigma_{\text{maximum}})$] of 0.1. The number of cycles

to cause complete failure or separation is taken as fatigue life (N_f).

3.4. Damage-failure analysis

Fracture surfaces of the deformed and failed test specimens were comprehensively examined in a JEOL scanning electron microscope (SEM) to both determine the macroscopic fracture mode as also characterize the fine-scale topography and microscopic mechanisms governing fracture. The distinction between the macroscopic mode and microscopic fracture mechanism is based on the magnification level at which the observations are made. In this context, the macroscopic mode refers to the nature of failure, while the microscopic mechanism relates essentially to the local failure processes (microscopic void formation, growth, coalescence and nature of cracking). Samples for SEM observation were obtained from the failed specimens by sectioning parallel to the fracture surface.

4. Results

4.1. Initial microstructure

The initial microstructure of the as-received 6061/SiC/10p-T6 and 6061/SiC/15p-T6 composites is shown in Fig. 1. The discontinuous SiC particulate (SiC_p) reinforcement phase, in the 6061 aluminum alloy matrix, were of non-uniform size, irregularly shaped and dispersed randomly through the alloy matrix. Few of the particulates were oblong or rectangular in shape and having sharp corners. At frequent intervals an agglomeration or clustering of the SiC_p reinforcements was observed resulting in particulate-rich and particulate-depleted regions (Fig. 1). An agglomerated site consisted of uniformly small and regularly shaped SiC_p intermingled with larger SiC_p of varying shapes and size. A non-uniform distribution of the reinforcing SiC particulates in the aluminum alloy matrix results in an anisotropic microstructure of the MMCs, i.e. 6061/SiC/10p-T6 and 6061/SiC/15p-T6. Besides the presence of reinforcing SiC_p , the alloy is rich in coarse insoluble intermetallic particles, which precipitate during ingot casting. These result from the presence of impurity elements, such as: iron and silicon, in the alloy. The coarse second-phase particles have been reported in earlier studies to be compounds of $\text{Al}_7\text{Cu}_2\text{Fe}$ and $\text{Al}_{12}(\text{FeMn})_3\text{Si}$ [47,48]. Their size ranging from 2 to 10 μm . The presence of grain refining element chromium results in the precipitation of dispersoids (Type: $\text{Al}_{12}\text{Mg}_2\text{Cr}$) during ingot preheat and high temperature homogenization treatments [47].

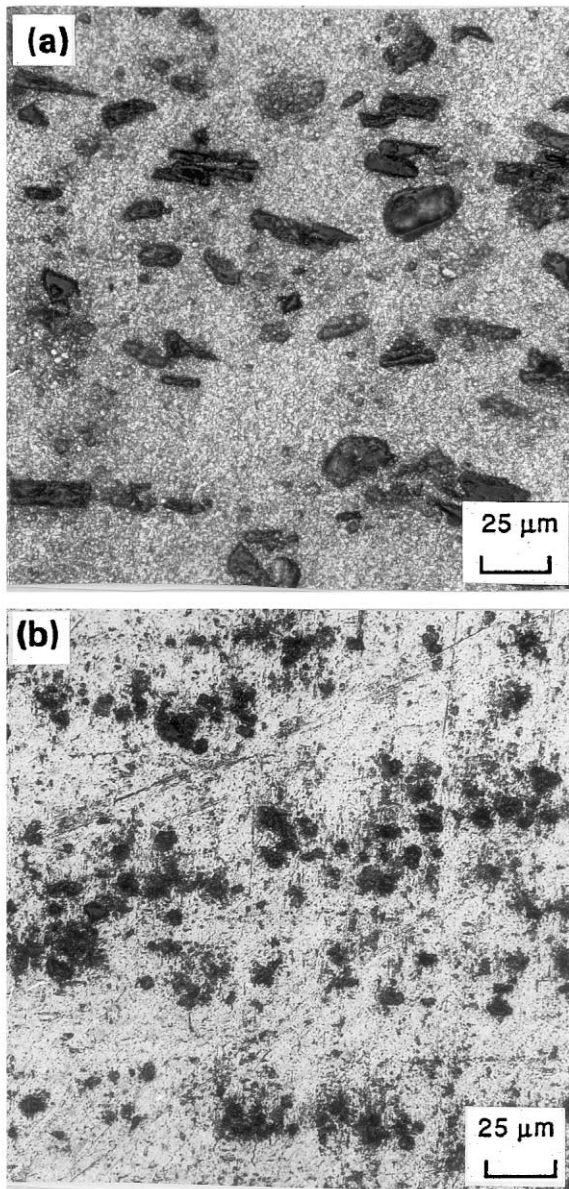


Fig. 1. Optical micrograph illustrating the microstructure of the 6061/SiC/xxp-T6 composites: (a) 10 vol.% of SiC_p and (b) 15 vol.% of SiC_p.

4.2. Tensile properties

The ambient temperature tensile properties of the

6061/SiC/xxp-T6 MMCs are summarized in Table 2. Duplicate samples were tested for each composite and no significant variation between the pairs of samples was observed. The ambient temperature yield strength, defined as the stress required at a plastic strain of 0.2%, of the discontinuously reinforced 6061 aluminum alloy metal matrix is marginally higher than that of the unreinforced aluminum alloy (6061-T6 = 278 MPa). However, the ultimate tensile strength (σ_{UTS}) of the discontinuously reinforced composite was only marginally greater than the tensile yield strength (σ_{YS}) providing sufficient evidence that the work hardening rate past yielding is low. In fact, the ultimate tensile strength followed the same trend as the yield strength of the 6061/SiC/xxp-T6 composite. The influence of increased volume fraction of SiC particulate reinforcement on improvement in strength of the 6061/SiC/xxp-T6 composite was at best marginal. However, for both volume fractions of the SiC_p reinforcement phase in the 6061 aluminum alloy metal matrix, the tensile ductility (strain-to-failure (ϵ_f)) was inferior to the unreinforced counterpart. The strain-to-failure decreased with an increase in volume fraction of the SiC particulate reinforcement in the metal matrix and is consistent with the improvement in strength and stiffness.

4.3. Tensile deformation and fracture

The tensile fracture surfaces are helpful in providing information relevant to developing an understanding of intrinsic microstructural effects on fracture properties of the 6061/SiC/xxp-T6 composites.

At the ambient temperature (27 °C) the 6061/SiC/xxp-T6 composites revealed limited ductility with fracture, on a macroscopic scale, occurring normal to the tensile stress axis. High magnification examination of the tensile fracture surfaces revealed features reminiscent of locally ductile and brittle mechanisms. Representative fractographs of the tensile fracture surface of the 6061/SiC/10p and 6061/SiC/15p composites are shown in Figs. 3 and 4.

Tensile fracture of the 6061/SiC/10p composite was relatively rough when viewed on a microscopic scale. Ductile regions described herein as tear ridges (Fig. 2a) surrounded the reinforcing SiC_p. High magnification observations revealed failure of the reinforcing SiC_p to

Table 2
Room temperature tensile properties of 6061 aluminum alloy/SiC composite

Reinforcement (%)	Temperature (°C)	Young's modulus (GPa)	Yield strength (Mpa)	Ultimate tensile strength (MPa)	Fracture stress (MPa)	Elongation to fracture (%)
0	27	66	287	316	145	26
10	27	81	278	332	313	16.3
15	27	89	343	372	353	7.0

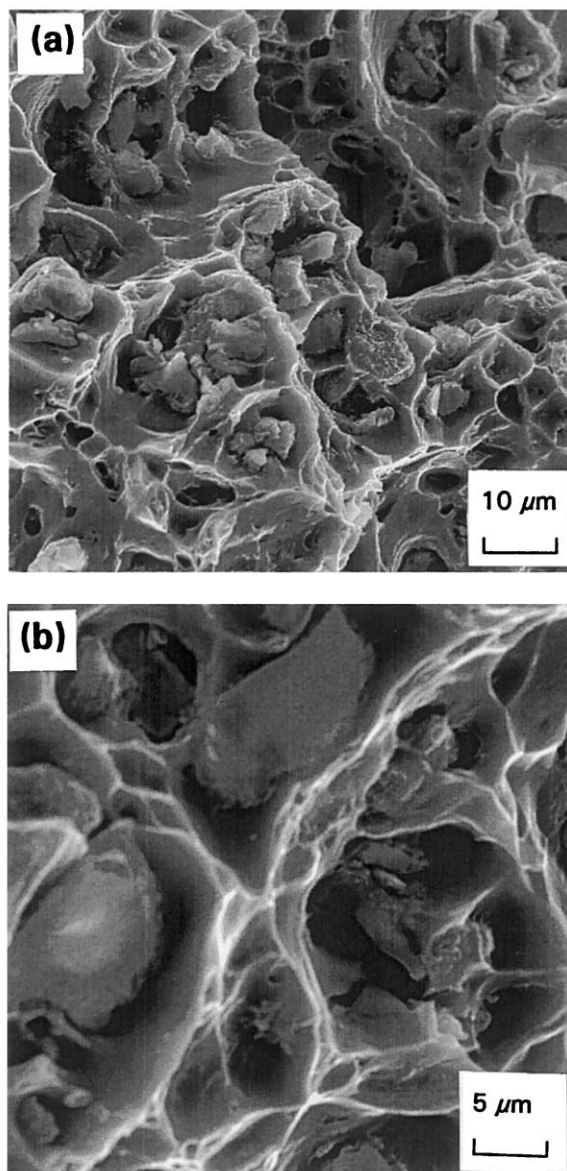


Fig. 2. Scanning electron micrographs of the tensile fracture surface of the 6061/SiC/10p-T6 composite deformed at 27 °C, showing: (a) tear ridges; (b) separation at matrix–particulate interfaces.

occur: (a) cracking (Fig. 2b); and (b) cracking and separation at the matrix–reinforcement interfaces (Fig. 2b). The matrix of the composite was covered with a population of fine microscopic voids. Few of the microscopic voids were intermingled with tear ridges and isolated pockets of dimple (Fig. 2b).

Tensile fracture of the 6061/SiC/15p composite was essentially flat and normal to the far-field stress axis when viewed on a macroscopic scale but rough when viewed on a microscopic scale. The matrix of the composite revealed combinations of: (a) tear ridges and microscopic voids (Fig. 3a); and (b) failure of the reinforcing SiC_p through cracking and separation at the

interfaces with the aluminum alloy matrix (Fig. 3b). The matrix of the composite was covered with a population of microscopic voids, of varying sizes and shapes. Two distinct types of dimples were observed: (a) those immediately adjacent to the reinforcing SiC_p; and (b) much smaller and shallower associated with coarse second-phase inclusions in the metal matrix. Multiple microscopic cracks were observed in regions of particulate agglomeration, which is partially responsible for observed degradation in tensile elongation to failure.

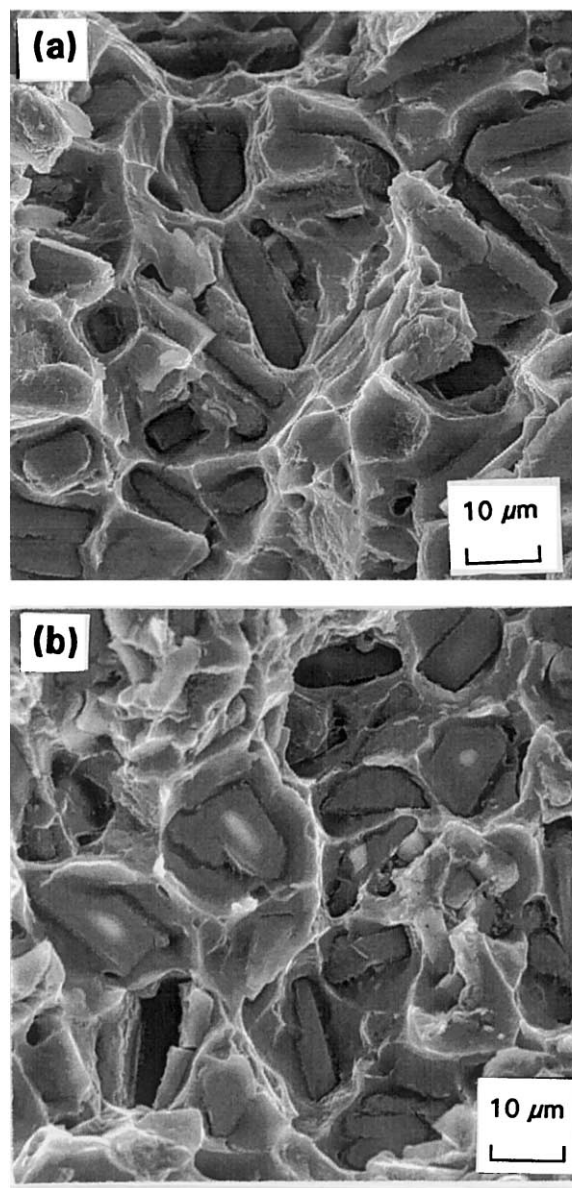


Fig. 3. Scanning electron micrographs of the tensile fracture surface of the 6061/SiC/15p-T6 composite deformed at 27 °C, showing: (a) tear ridges and voids; (b) cracking of the particulates and interface decohesion.

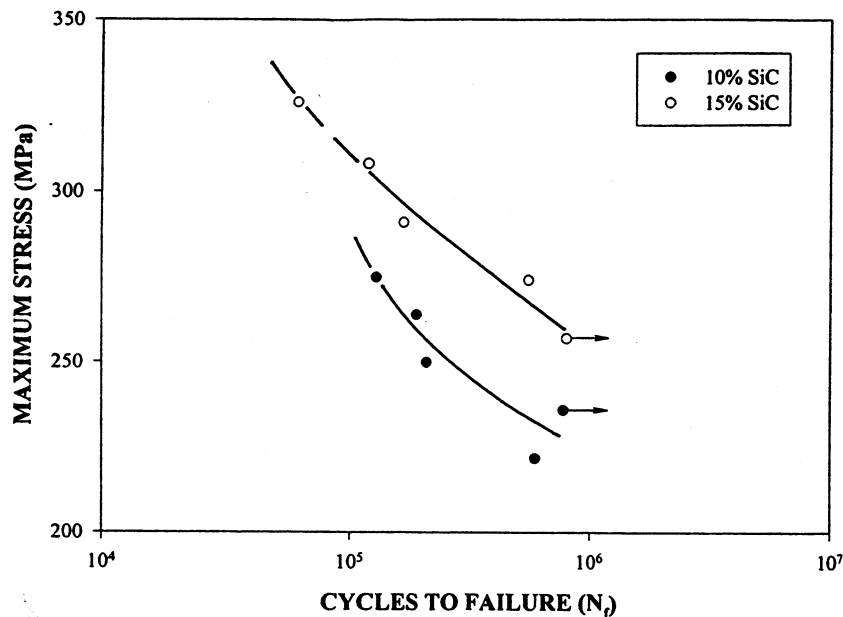


Fig. 4. Influence of reinforcement volume fraction on the maximum stress vs. fatigue life for the 6061/SiC/xxp-T6 composites.

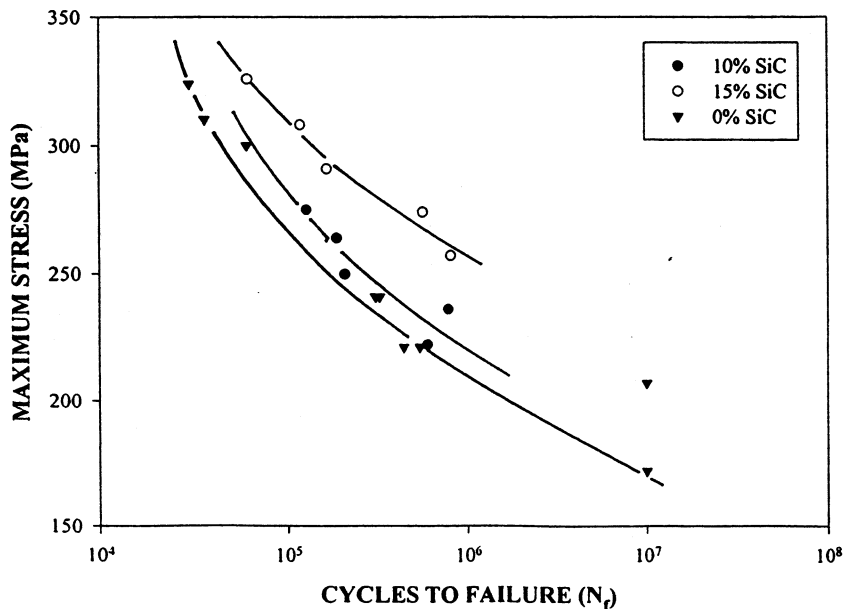


Fig. 5. Comparing the maximum stress vs. fatigue life response of the 6061/SiC/xxp-T6 composites with the unreinforced alloy (6061-T6).

4.4. Cyclic stress resistance: high cycle fatigue

The purpose is to establish an endurance limit or fatigue limit, below which stress the material can be cycled infinitely. The endurance limit for this aluminum alloy-based composite material is taken at 10^6 cycles. In high cycle-low stress fatigue of the discontinuously-reinforced 6061 aluminum alloy composite material the property changes are essentially confined to occur in the fracture process zone at the crack tip, while the material in the far-field is not appreciably affected during fatigue cycling. The results shown in Fig. 4, as the variation of maximum

stress [σ_{maximum}] as a function of number of cycles-to-failure (N_f) reveal the 6061/SiC/xxp-T6 composite material to follow a pattern quite similar and expected of monolithic aluminum alloys and their composite counterparts. At ambient temperature the endurance stress of the 6061/SiC/15p-T6 composite is 270 MPa and that of the 6061/SiC/10p-T6 composite is 240 MPa. A comparison of maximum stress versus cyclic fatigue life of the 6061/SiC/xxp-T6 composites with the unreinforced matrix (6061-T6) is made in Fig. 5. The presence of SiC particulate reinforcements in the metal matrix has increased the endurance limit from 210 to 270 MPa.

The beneficial effect of increasing SiC_p reinforcement volume fraction, for a constant particulate size, was observed on cyclic fatigue life as well. In Fig. 5 is compared the maximum stress versus cycles-to-failure for the unreinforced and composite materials with varying reinforcement volume fractions. Increasing the SiC_p volume fraction resulted in higher fatigue strength (taken as the highest stress at which a specimen survived 10^6 cycles). In particular, the highest volume fraction (15%: 6061/SiC/15p-T6) showed noticeable improvement in fatigue strength compared to the unreinforced matrix (6061-T6). The fatigue ratio, defined herein as the ratio of fatigue strength to yield strength, was fairly constant for the two composites and higher than that of the unreinforced 6061-T6-aluminum alloy. The high strength of the 6061/SiC/xxp-T6 MMCs compared to the unreinforced 6061-T6 metal matrix is responsible for the superior high-cycle fatigue resistance, that is, cyclic fatigue life of the composites. However, increasing the SiC_p reinforcement content resulted in an increased elastic modulus of the composite. Thus, at equivalent stress amplitudes the composite with higher SiC reinforcement content experiences lower cyclic strain as shown by Vaidya and Lewandowski [34]. If variation of elastic strain, instead of maximum stress, with number of cycles-to-failure is examined, the difference in cyclic fatigue response between the reinforced and unreinforced alloys becomes apparent (Fig. 6). In this curve, reinforcing with SiC particulates resulted in a reduction of failure cycles by about two orders of magnitude. Comparing the reinforced alloys reveals little to no influence on cyclic fatigue life with an increase in SiC_p reinforcement con-

tent from 10 to 15-vol.%. The lower fatigue lives of the reinforced materials is attributed to combination of a relatively short crack initiation period in comparison with the unreinforced counterpart (6061-T6) and enhanced crack propagation rates, details of which are discussed in the following section.

4.5. Cyclic deformation and fracture

An examination of the cyclic fatigue fracture surfaces of the deformed and failed 6061/SiC/xxp-T6 composite specimens was made in a JEOL scanning electron microscope (SEM) at:

1. low magnifications with the purpose of identifying the regions of: (a) microscopic crack initiation and stable crack growth, and (b) final fracture or overload, and
2. higher magnifications in: (a) region of early microscopic crack growth to identify the size, location and number of microscopic cracks and their early progression through the composite microstructure; and (b) region of overload to identify the fine-scale features reminiscent of local governing mechanisms.

For a given volume fraction of the SiC_p reinforcement phase in the 6061 aluminum alloy matrix the fatigue fracture surfaces revealed near similar topographies at all cyclic stress amplitudes. However, on a microscopic scale, the fatigue fracture features were found to vary with particulate reinforcement content in the aluminum alloy (6061) matrix. Representative fractographs of the fatigue fracture surfaces, for the two DRAMMCs, is shown in Figs. 7–11. These are discussed in detail below.

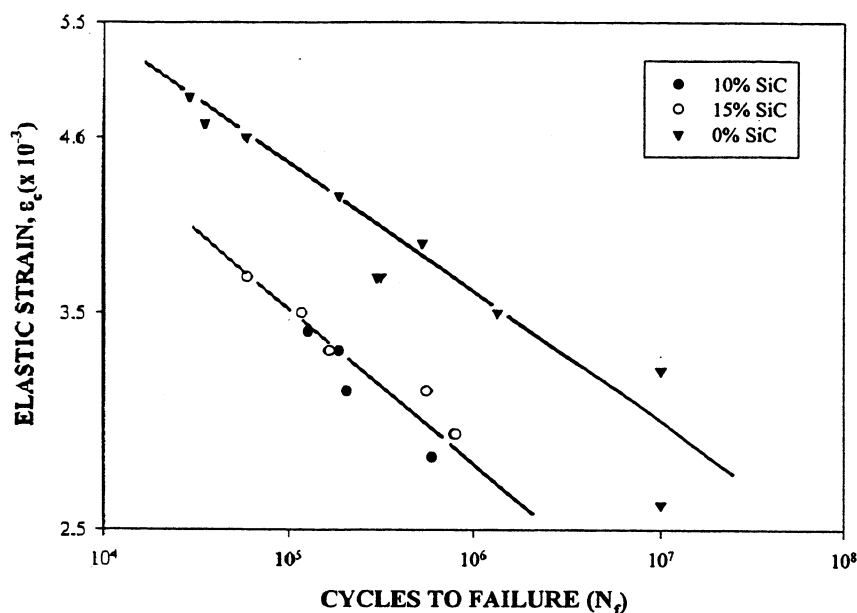


Fig. 6. Comparing the cyclic strain (elastic) amplitude with fatigue life (number of cycles to failure) of the 6061/SiC/xxp-T6 composites with the unreinforced alloy (6061-T6).

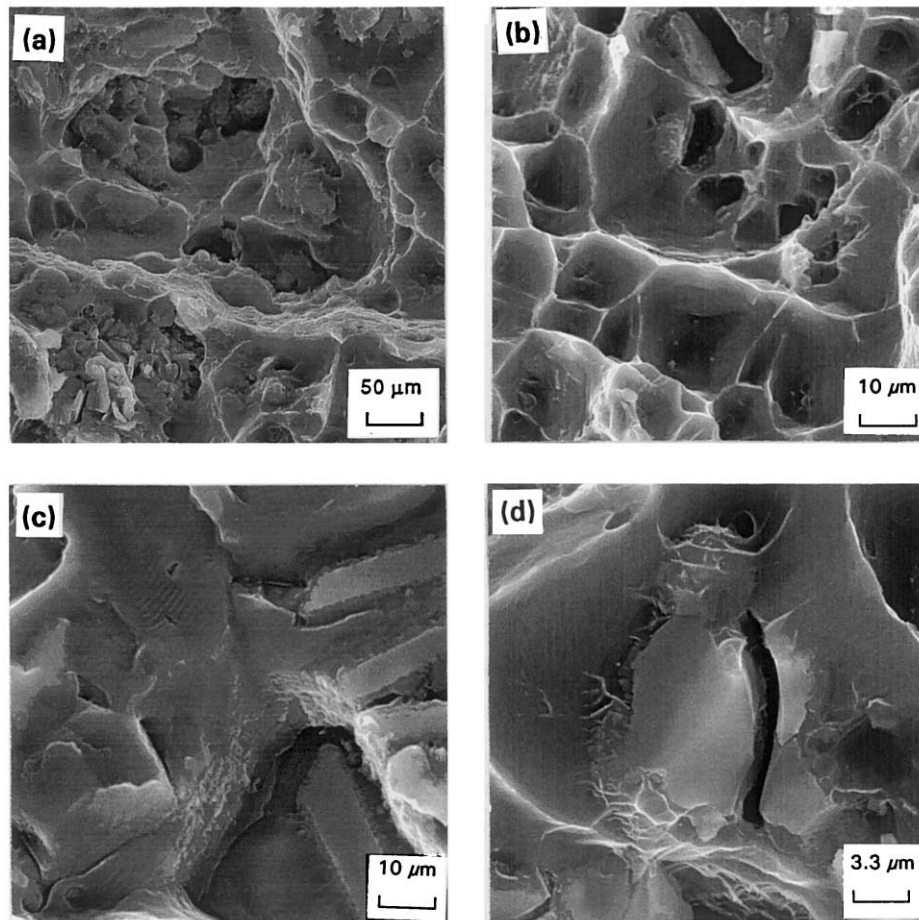


Fig. 7. Scanning electron micrographs of the 6061/SiC/10p-T6 composite; cyclically deformed at maximum stress of 0.99 yield strength (N_f : 276 640 cycles), showing: (a) tear ridges; (b) microscopic voids; (c) striations; (d) microscopic crack and cracked particulates.

4.5.1. 6061/SiC/10p-T6 composite

At the higher cyclic stress amplitudes the fracture behavior of this composite was essentially normal to the far-field stress axis. Macroscopic, or low magnification observations, revealed the fracture surface to essentially comprise features reminiscent of brittle (microscopic cracking) and locally ductile mechanisms. The region of stable and early microscopic crack growth was overall very small at high cyclic stress amplitudes and concomitant short fatigue life (Fig. 7), and at the lower cyclic stress amplitudes and resultant enhanced fatigue life (Fig. 8). The region of stable microscopic crack growth was essentially flat and contained numerous microscopic cracks, tear ridges (Fig. 7a), a population of voids of varying shapes (Fig. 7b) and isolated pockets of striations (Fig. 7c). The microscopic cracks were found in large numbers in the sample deformed to failure at the lower cyclic stress amplitude ($\sigma_{\max} = 0.85\% \sigma_{\text{yield}}$) and resultant enhanced fatigue life (Fig. 8b). The region of transition from stable to rapid unstable microscopic crack growth and overload failure was not distinctly evident (Fig. 8a). Present in the region of early microscopic crack growth

were: (a) tear ridges surrounded by shallow dimples (Fig. 8c); (b) failure by separation at the matrix-reinforcement (SiC_p) interfaces (Fig. 8d); and (c) cracked SiC_p along with other coarse second-phase particles in the microstructure. The microscopic voids, of varying size, were distributed randomly through the fracture surface. The macroscopic and fine microscopic voids were: (a) surrounded by cracked SiC_p and second-phase particles; and (b) isolated regions of shallow dimples. During stress amplitude-controlled deformation the larger SiC_p tended to definitely fail by cracking (Fig. 9). Besides, at higher stress levels the larger SiC_p and particulate clusters are favored sites since they act as larger stress concentration sites when compared to the smaller SiC_p counterparts.

4.5.2. 6061/SiC/15p-T6 composite

Fatigue failure of this composite material was essentially similar at both high cyclic stress and low cyclic stress amplitudes. The region of stable microscopic crack growth was small. A large portion of the fatigue fracture surface revealed evidence of: (a) microscopic cracking throughout the composite microstructure (Fig.

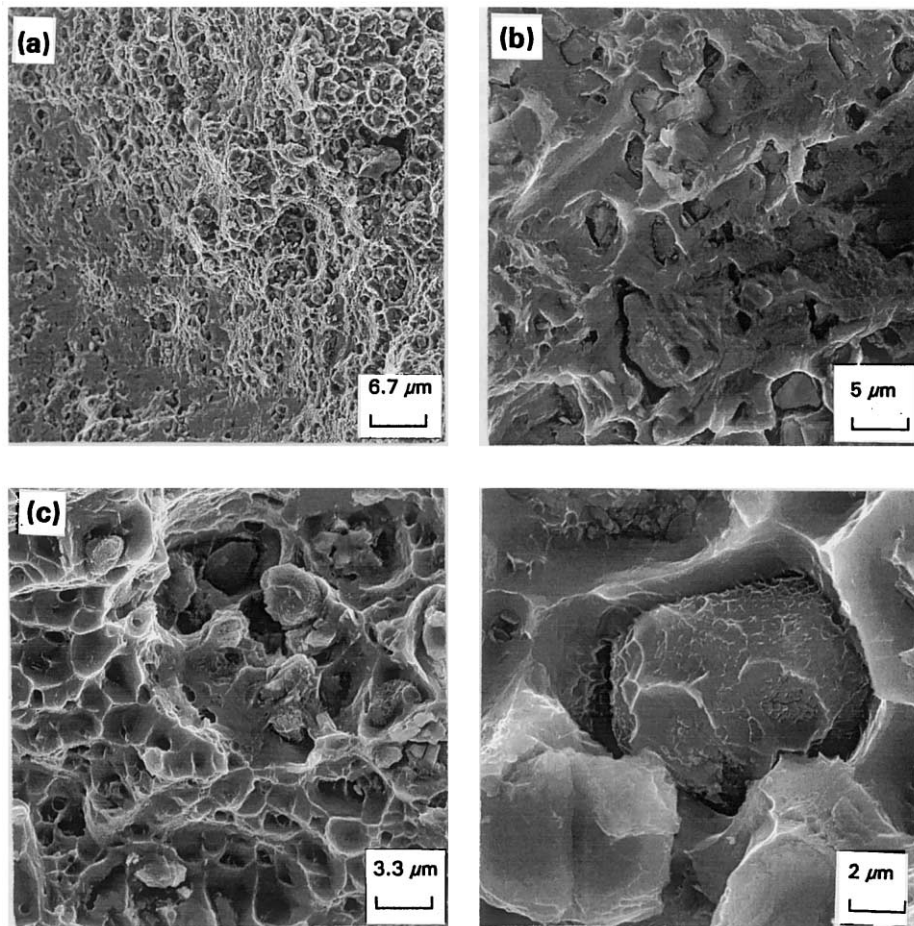


Fig. 8. Scanning electron micrographs of the 6061/SiC/10p-T6 composite; cyclically deformed at maximum stress of 0.85 yield strength (N_f : 780 799 cycles), showing: (a) overall morphology; (b) microscopic crack; (c) microscopic voids and shallow dimples; (d) particle clustering and decohesion at the interfaces.

10a); (b) failure of the reinforcing SiC_p through cracking and tear ridges (Fig. 10b); and (c) cracking and separation at the particulate reinforcement-matrix interfaces and the ensuing development of microscopic cracks (Fig. 10c). The matrix revealed pockets of fine striation-like features reminiscent of stable microscopic crack growth (Fig. 10d). High magnification observations of the sample deformed at low cyclic stress amplitude and resultant enhanced fatigue life revealed the following:

1. An overall rough fracture surface (Fig. 11a).
2. Evidence of gross brittle failure through failure of a large number of the reinforcing SiC particulates by both cracking and decohesion at interfaces with the matrix (Fig. 11b).
3. Isolated pockets of striation-like features (Fig. 11c).
4. Traces of ductile failure through the formation and presence of shallow dimples surrounding the cracked SiC particulates (Fig. 11d).

A large mismatch in strain carrying capability exists between the intrinsically brittle and elastically deforming reinforcing SiC_p and the soft and plastically deforming aluminum alloy metal matrix. This results in a

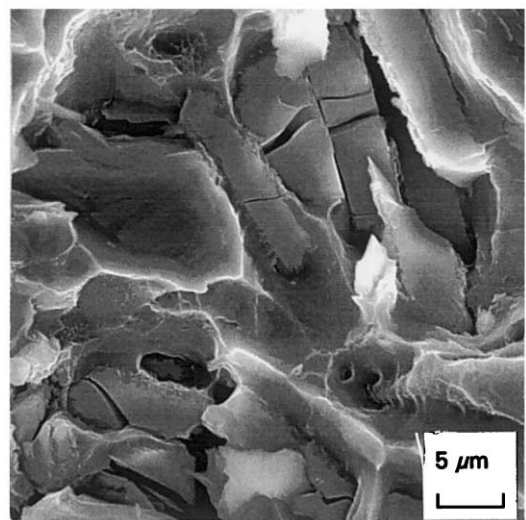


Fig. 9. Scanning electron micrograph of the 6061/SiC/15p-T6 composite showing microcrack initiation at region of the particulate agglomeration.

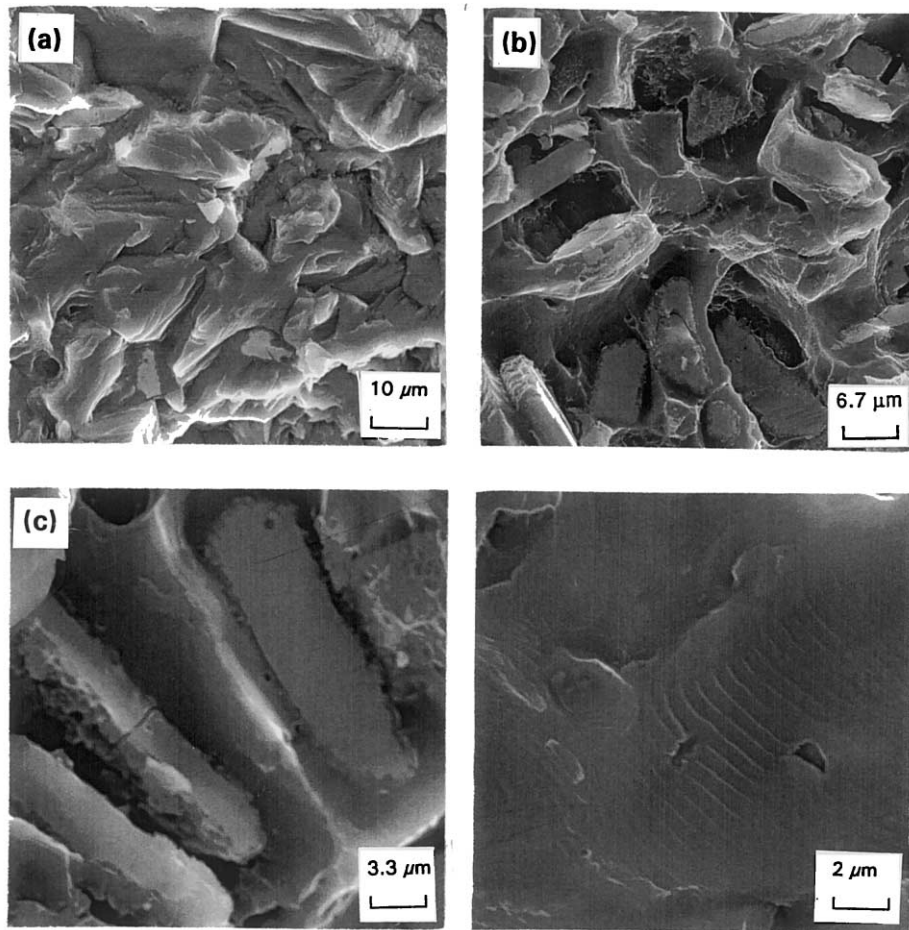


Fig. 10. Scanning electron micrographs of the 6061/SiC/15p-T6 composite; cyclically deformed at maximum stress of 0.75 yield stress (N_f : 806 291 cycles), showing: (a) overall morphology at high magnification; (b) tear ridges and particulate clusters; (c) cracked particulate and failure at the interface; (d) fine striations on the transgranular surface.

large concentration of stress near the reinforcing SiC_p causing them to crack and the adjacent matrix to fail by ductile fracture. Failure through the conjoint influences of separation at the matrix-reinforcement particulate interfaces, referred to in this manuscript as decohesion, and cracking of the reinforcing SiC_p suggests the exacerbation of microplastic deformation in regions containing a high volume fraction of the reinforcing particulates (SiC_p).

5. Discussion

The constraints in mechanical deformation arising as a result of the hard, brittle and elastically deforming SiC particulates in a soft, ductile and plastically deforming aluminum alloy metal matrix and the concomitant development of a triaxial stress state aids in limiting the overall flow stress of the composite matrix. This favors void initiation and growth in the matrix, and failure by both cracking of the SiC_p and separation at the particulate–matrix interfaces. Furthermore, as a

direct consequence of the deformation constraints a higher applied maximum stress is required to initiate microplastic deformation in the composite. This translates to a higher elastic constant and higher yield strength of the composite material. Under the influence of a far-field tensile load the microscopic voids appeared to have undergone limited growth confirming a possible contribution from reinforcement particulate (SiC_p) constraint-induced triaxiality on failure of the composite microstructure. Failure of the SiC particulate(s) is governed by the competing influences of: (1) local plastic constraints; (2) particle size; and (3) degree of agglomeration. The local plastic constraints are particularly important for the larger-sized particles and particle clusters during composite fracture [49–51].

Examination of the fracture surfaces revealed damage to be localized at the reinforcing phase through cracked particulates and failure by cracking and separation at interfaces with the matrix, with little evidence of microscopic void formation away from the cracked particulate. This suggests that microplastic deformation become localized during the early stages of cyclic defor-

mation in the high cycle regime. The intrinsic brittleness of the reinforcing SiC_p coupled with the propensity for it to fracture due to localized deformation results in both cracking and interfacial failure being the dominant damage mode. The higher yield strength coupled with damage to the composite microstructure arising from the conjoint influence of particulate cracking and interfacial failure is responsible for the inferior ductility of the (6061/ $\text{SiC}/\text{xyp-T6}$) MMCs over the unreinforced alloy (6061-T6). Furthermore, neglecting any contribution from the interfaces, the triaxial stresses generated during far-field loading favor limited growth of the microscopic voids in the matrix of the composite. The limited growth of the microscopic voids coupled with lack of their coalescence, as a dominant fracture mode for the 6061/ $\text{SiC}/\text{xyp-T6}$ MMCs, clearly indicates that the deformation properties of the aluminum alloy matrix are significantly altered by the presence of the discontinuous SiC particulate reinforcements. In SiC particulate-rich regions of the matrix, fracture occurred early and the damage propagated rapidly among the particulate agglomerate. However, the reinforcement-

lean region, i.e. the metal matrix, aids in retarding the progression and linkage of damage.

Fracture of the brittle SiC_p exacerbates the local damage and serves as a starting point for the formation of a microscopic crack. This in conjunction with failure of the surrounding matrix results in the formation of fine microscopic voids at the particulate–matrix interfaces. Very few of the fine microscopic voids coalesce and the halves of these voids are the shallow dimples observed surrounding the cracked particles. The lack of formation of ductile dimples in the matrix, as a dominant fracture mode, is attributed to the constraints on plastic flow caused by the presence of SiC_p , i.e. the deformation incompatibility between the plastically deforming metal matrix and the elastically deforming reinforcement phase, and not due to limited ductility of the aluminum alloy (6061-T6) metal matrix. With an increase in reinforcement content, microscopic fracture was dominated by cracking of the particulates on account of their intrinsic brittleness. With a progressive increase in the maximum applied stress, the larger-sized particles fracture first, at low values of ‘microplastic’

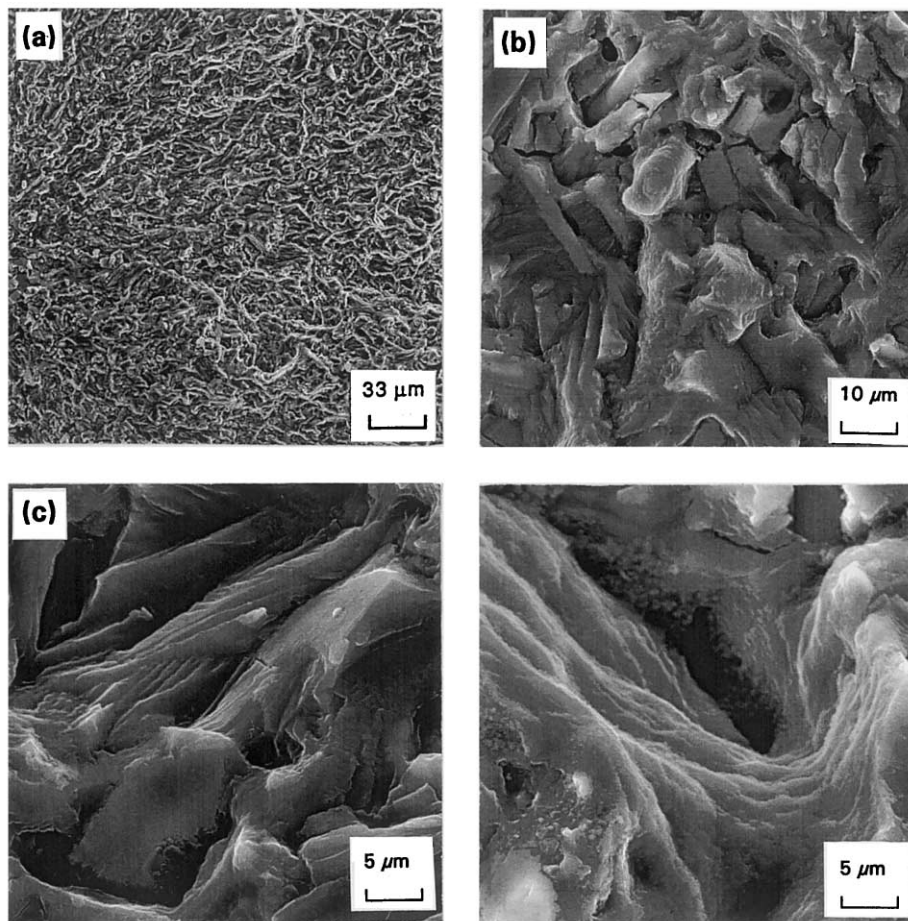


Fig. 11. Scanning electron micrographs of the 6061/ $\text{SiC}/15\text{p-T6}$ composite; cyclically deformed at maximum stress of 0.80 yield stress ($N_f = 559\,755$ cycles), showing: (a) overall morphology; (b) cracking of the reinforcing particulates; (c) pockets of striations; (d) shallow dimples surrounding cracked particulate.

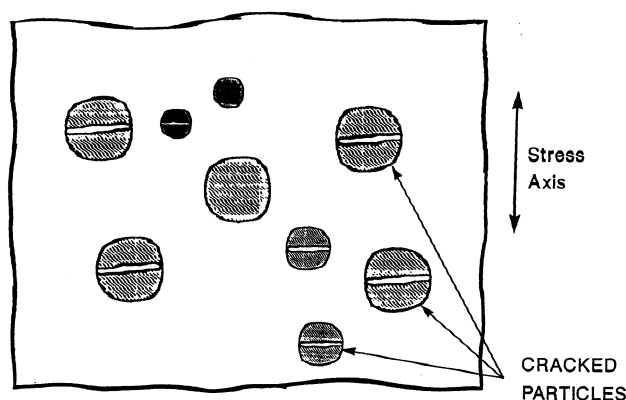


Fig. 12. Schematic showing fracture of the particulates dispersed in the 6061 alloy matrix in a direction normal to the applied stress.

strain, followed by the smaller-sized particles. Also, tensile (quasi-static) fracture surfaces revealed the fracture plane of cracked particles to be essentially normal to the far-field stress, suggesting importance of normal stress in inducing particle fracture (Fig. 11). The overall ‘damage’ resulting from mechanical deformation of the 6061/SiC/xp MMC can be associated with failure of the discontinuous reinforcement through: (a) cracking; and (b) separation at interfaces with the matrix. The broken SiC particulates elevate the local stress, which raises the probability of failure of the matrix and other particulates and second-phase particles in the immediate vicinity. Final fracture is achieved by damage propagation through the matrix between particle-rich regions. The macroscopic voids were intermingled with fine microscopic voids. Few of the voids generated by particle cracking did not grow extensively in the normal stress direction, which is generally the case in ductile fracture of unreinforced aluminum alloys. The lack of extensive void growth in this SiC particulate-reinforced 6061 aluminum alloy metal matrix also suggests that the fracture strain is critically controlled by void nucleation strain and associated linkage strain. The more widely separated macroscopic voids were found to coalesce by linking of the microscopic voids to form void sheets (Fig. 12).

An observation of the quasi-static and cyclic fatigue fracture surfaces of the two chosen composites revealed that excess of 50% of the reinforcing particulates had fractured during deformation. This indicates that not all of the reinforcing particulates were loaded to their fracture stress suggesting the non-uniform distribution of particulates in the metal matrix. This situation is exacerbated by the mismatch strain and concomitant high compression stresses in the composite matrix due to differences in the thermal expansion coefficient between constituents of the composite; the SiC_p and the 6061-T6 aluminum alloy matrix. For the reinforcing SiC_p to fracture the applied far-field tensile stress will

have to first overcome the internal compressive stress present in the particulate.

6. Conclusions

The influence of volume fraction of discontinuous SiC particulate reinforcement on cyclic stress response, fatigue life and final fracture behavior of 6061-T6 aluminum alloy-based composite provides the following key observations.

The as-received microstructure of the 6061/SiC-T6 composites revealed a non-uniform distribution of the reinforcing particulates, of non-uniform size, along the extrusion direction of the composite plate. At regular intervals, an agglomeration or clustering of the SiC_p, of varying size, was observed.

The non-uniform size and dispersion of the reinforcing phase caused the particles to crack at low values of applied stress. The cracks initiated both at and near the particulate–matrix interfaces and in regions of particulate agglomeration. The quasi-static fracture surfaces revealed limited ductility or brittle appearance on a macroscopic scale, but at the microscopic level features reminiscent of locally ductile and brittle rupture mechanisms. The fracture surface revealed combinations of tear ridges; cracked particulates and separation through decohesion at the matrix–particulate interfaces.

Increasing the SiC_p volume fraction resulted in higher fatigue strength (taken as the highest stress at which a specimen survived 10⁶ cycles). In particular, the highest volume fraction (15%: 6061/SiC/15p-T6) showed noticeable improvement in fatigue strength compared to the unreinforced matrix (6061-T6). The high strength of the 6061/SiC/xp-T6 MMCs compared to the unreinforced 6061-T6 metal matrix is responsible for the superior high-cycle fatigue resistance of the composites.

For a given volume fraction of the SiC particulate reinforcement phase in the 6061 aluminum alloy metal matrix, fracture morphology was observed to be essentially similar at the different cyclic stress amplitudes. Macroscopic observations revealed the fracture surface to be brittle, that is: low cyclic ductility but microscopically features reminiscent of locally ductile (voids and dimples) and brittle (microscopic and macroscopic cracks) mechanisms. With an increase in SiC_p contents in the metal matrix, fractures of the composite were dominated by particulate cracking and decohesion at the particulate–matrix interfaces.

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